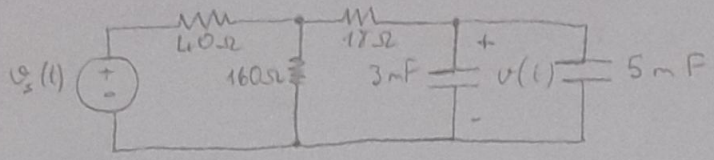
ELEC 225 ASSIGNMENT-1

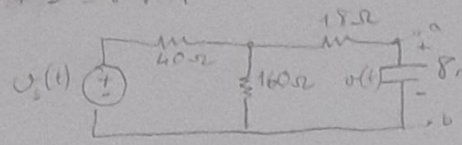
Q1) The voltage source in the circuit shown in Figure 1 is

$$v_s(t) = 4 + 6u(t)\ V$$

Determine $v(t)$ for $t > 0$



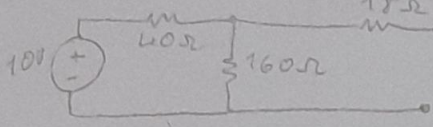
$C_p = 3 + 5 = 8\ mF$



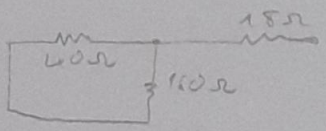
$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

Let's apply thevenin equivalent

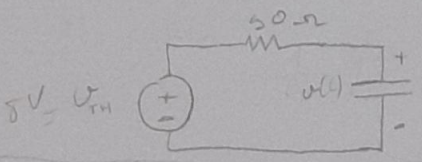
$v_s(t) = 10\ V \quad [t > 0]$



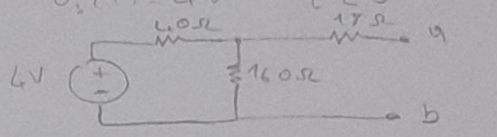
$$V_{ab} = V_{oc} = \frac{160}{160+40}\ 10 = 8\ V$$



$$R_{th} = 40 \,//\, 160 + 18 = \frac{40.160}{40+160} + 18 = 50\Omega$$

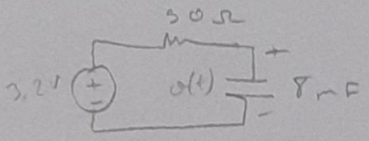


$v_s(t) = 4V \quad t < 0$



$$V_{ab} = V_{oc} = \frac{160}{200} \cdot 4 = 3.2\ V$$

$R_{th} = 50\Omega$



$$v(t) = v(\infty) + (v(0) - v(\infty))\, e^{-t/\tau}$$

$\tau = RC = 400$

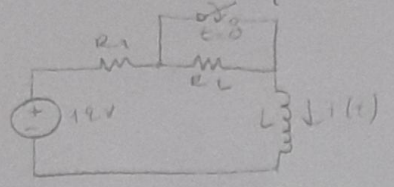
$$v(t) = 8 - 4.8 \cdot e^{-t/400}$$

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Q2) The circuit shown in Figure-2 is at steady state before the switch closes at time $t=0$. After the switch closes the inductor current is given by

$$i(t) = \underbrace{0,3}_{\text{forced}} - 0,2e^{-5t} \text{ A} \quad \text{for } t \geq 0$$

Determine the values of R_1, R_2 and L



$$S = \frac{1}{\tau} \qquad \tau = \frac{L}{R} \qquad t>0 \Rightarrow R_1 = 5L$$

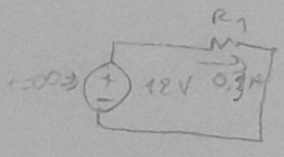
$$i(\infty) = 0,3 \qquad S = \frac{R}{L}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]\, e^{-t/\tau}$$

$$[i(0) - i(\infty)] = -0,2$$

$$i(0) - 0,3 = -0,2$$

$$\boxed{i(0) = 0,1 \text{ A}}$$



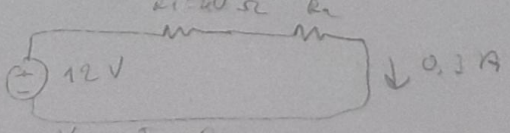
$$V = I \cdot R$$

$$12 = 0,3 \cdot R_1$$

$$\boxed{R_1 = 40\ \Omega}$$

$$L = \frac{R_1}{5} = 8 \text{ H}$$

$(t=0) \rightarrow i = 0,1$ A



$$V = I \cdot R_{eq}$$

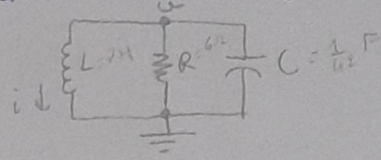
$$12 = 0,1 \cdot (40 + R_2)$$

$$12 = 4 + 0,1R_2$$

$$\boxed{R_2 = 80\ \Omega}$$

Fouad Aladham

Q3) Find the natural response of the RLC circuit of the Figure-3 when $R = 6\,\Omega$, $L = 7\,H$, and $C = 1/42\,F$. The initial conditions are $v(0) = 0$ and $i(0) = 5A$



First write the characteristic equation:

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$

$$s^2 + \frac{1}{\frac{6}{42}}s + \frac{1}{\frac{7}{42}} = 0$$

$$s^2 + 7s + 6 = 0$$

$$(s+6)(s+1) = 0$$

$$s_1 = -6 \quad s_2 = -1$$

$$\frac{v(0)}{R} + i(0) + C\frac{dv(0)}{dt} = 0$$

$$\frac{dv(0)}{dt} = -\frac{v(0)}{RC} - \frac{i(0)}{C}$$

the natural response is $v_n = A_1 e^{-6t} + A_2 e^{-t}$

$$v_n(0) = A_1 + A_2$$

$$A_1 + A_2 = 0$$

$$\frac{dv_n(0)}{dt} = s_1 A_1 + s_2 A_2 = -\frac{v(0)}{RC} - \frac{i(0)}{C}$$

$$-6A_1 - A_2 = -210$$

$$A_1 = -A_2 \qquad 6A_2 - A_2 = 5A_2 = -210$$

$$\boxed{A_2 = -42}$$

$$\boxed{A_1 = 42}$$

$$v_n = (+42e^{-6t} - 42e^{-t})\ V$$

Fouad Hadhami